CHING-WEI (WILLIAM) KANG

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EDUCATION

University of Wisconsin-Madison **Madison, WI**
B.S. Computer Science and Data Science | GPA: 3.75/4.0 Expected May 2027
Relevant Coursework: Software Engineering, Artificial Intelligence, Data Management for Data Science

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, Go, C++
Web / Backend: React, Next.js, Node.js, FastAPI, Express, PostgreSQL, Redis, SQLite, REST APIs
Infrastructure / Delivery: Docker, AWS, RBAC, SSE, CI/CD, automated testing, observability

OPEN SOURCE ENGINEERING

204 external PRs merged across 64 repos; 37 at Microsoft, NVIDIA, Meta, Cloudflare, Docker, and GitHub.

- Microsoft APM PR #1993 — Reused cached directory analysis to cut **500 matcher calls to 0** in a 500-directory benchmark while preserving **instruction placement and root behavior** across **1,206 compilation tests**.
- Cloudflare vinext PR #2521 — Propagated Next.js **serverExternalPackages** into Nitro **traceDeps without duplication**, preserving PostgreSQL and other **server-only dependencies** in **production bundles** with regression coverage.
- Microsoft APM PR #1971 — **Canonicalized deny-listed packages** across casing, `.git` suffixes, and GitHub URLs to **block policy bypasses**, then restricted deployed POSIX binaries to **owner-only 0700 permissions**.
- NVIDIA NeMo Agent Toolkit PR #2100 — Added **paired LangChain start/end events** with **run IDs, tags, timing, inputs/outputs**, and cleanup on success/error, preventing **stale trace state** and validating complete agent lifecycles across **684 package tests**.
- Meta DotSlash PR #16 — Verified GitHub release assets with **SHA-256** before extraction across **macOS, Windows, ARM64, and x86-64**, rejecting **missing or mismatched digests** before installation and closing a **cross-platform supply-chain gap**.
- Docker CLI PR #6957 — Routed image-push notes through Docker's **shared TUI output** and added **NO_COLOR regression coverage**, removing **ANSI escapes** and keeping **accessible logs, terminals, and CI automation deterministic**.
- NVIDIA NemoClaw PR #6084 — Added **verified user-local shims** for **all 3 CLI aliases** across **POSIX and Windows**, restoring documented commands while preserving failures and keeping **secondary aliases best-effort**.
- hflow PR #184 — Implemented **lossless H.264 AUD repair** for ROS2 CDR/protobuf streams, distinguishing **multi-slice** from **invalid multi-picture data** while preserving conforming payloads **byte-for-byte** across **997 tests**.

PROFESSIONAL EXPERIENCE

Full-Stack Engineering Intern | Global AI (Global API Inc.) **New York, NY | Hybrid | May 2026–Present**

- **Led Python/FastAPI backend development** for an AI video pilot producing **~50 videos/day**; orchestrated parallel jobs with PostgreSQL/Redis, cutting turnaround from **20+ to ~12 minutes**.
- **Cut per-video model cost by 70–80%** through staged execution, cached Gemini outputs, reusable intermediate assets, and model routing that avoided repeated generation calls.
- **Raised completion from 90% on first attempt to 99% after retries** using checkpoints, idempotent retries, and resumable execution, recovering **90% of initially failed jobs**.
- **Shipped a Next.js/TypeScript dashboard with SSE progress streaming** and per-stage reruns; partnered with the CEO and **4 departments** to translate operating needs into product workflows.

AI & SWE Intern | Guangdong Akang Health Technology Group **China | On-site | Jul 2026–Aug 2026**

- **Built retrieval interfaces and JSON/SQLite pipelines** ingesting **1M+ web documents from 30+ feeds** and standardizing **4.5M+ records** for reusable pharmaceutical, legal, and company analysis.
- **Automated data-quality checks** for completeness, citations, and source coverage; captured source IDs, URLs, retrieval dates, and validation metadata for reproducible review.
- **Designed multi-agent research workflows** with evidence ranking and human-review checkpoints, keeping pharmaceutical findings traceable and reviewable before internal use.
- **Delivered multilingual translation and fine-tuning workflows** for internal WeChat tools and taught AI fundamentals to **100+ managers**, supporting adoption and reuse of domain research.

SELECTED PROJECTS

ClearMove — Rental Evidence Platform | **Oct 2025–Present**
https://devpost.com/software/clearmovem

- **Built a Next.js/React/TypeScript platform spanning 21 API routes and 10 PostgreSQL tables**, integrating onboarding, lease/checklist parsing, validation, and timestamped PDF reports for stakeholder review.
- **Implemented token-based workflows and auditable exports**, turning inspection and lease inputs into reproducible, shareable records for stakeholder review and downstream reporting.